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Project: Database

Hospital Management System

1. Introduction

The Hospital Management System is a database project designed to manage hospital operations efficiently. The system stores patient information, doctor records, appointments, departments, billing, and treatment details. The project demonstrates concepts of relational database systems using SQL.

2. Project Objective

The main objectives of this project are:

- Manage patient records
- Manage doctor and department information
- Schedule appointments
- Store billing records
- Generate reports using SQL queries
- Demonstrate database normalization and relationships

3. Scope of the System

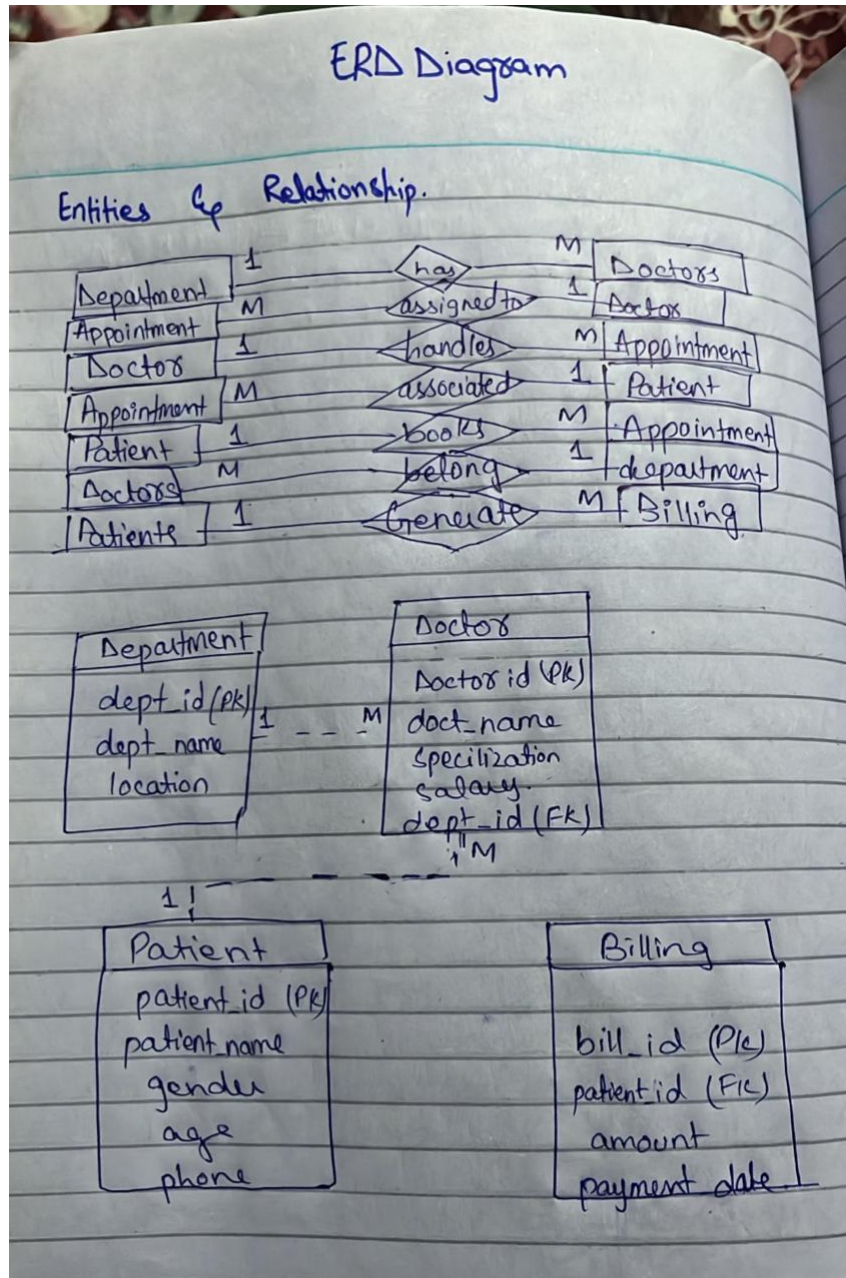
The Hospital Management System can:

- Add new patients
- Add doctors and departments
- Schedule appointments
- Generate bills
- Store treatment information

- Display reports using joins and aggregate functions

4. ERD (Entity Relationship Diagram)

The following ERD represents the relationship between all entities in the Hospital Management System database.



5. Database Design

The database consists of the following tables:

1. Department
2. Doctor
3. Patient
4. Appointment
5. Billing

The database uses:

- Primary Keys
- Foreign Keys
- Constraints
- Relationships
- Views

6. Primary Keys and Foreign Keys

Table	Primary Key	Foreign Key
Department	dept_id	None
Doctor	doctor_id	dept_id
Patient	patient_id	None
Appointment	app_id	patient_id, doctor_id
Billing	bill_id	patient_id

7. Constraints

Primary Key Constraints

- dept_id

- doctor_id
- patient_id
- app_id
- bill_id

Foreign Key Constraints

- dept_id references Department
- patient_id references Patient
- doctor_id references Doctor

NOT NULL Constraints

- doctor_name
- patient_name
- dept_name

8. Relationships

Relationship	Type
Department → Doctor	One-to-Many
Doctor → Appointment	One-to-Many
Patient → Appointment	One-to-Many
Patient → Billing	One-to-Many

9. Normalization

First Normal Form (1NF)

- All attributes contain atomic values.
- No repeating groups.

Second Normal Form (2NF)

- Partial dependency removed.
- Every non-key attribute depends on the primary key.

Third Normal Form (3NF)

- Transitive dependency removed.
- Non-key attributes depend only on primary keys.

10. GitHub Repository

GitHub Repository Link:

<https://github.com/muhammadmuneebmanzoor/Hospital-Management-System-DB.git>